

Backend Developer Take-Home Coding Challenge

Wii Group

Role: Backend Developer

Time Limit: 3-4 hours (can be completed over 2 days)

Challenge: Provider Registration API

Build a simple REST API for provider registration with the following features:

Required Endpoints

1. POST /api/providers/register

Function: Register a new service provider

Input: - name (string, required) - email (string, required, valid email format) - phone (string, required) - business_type (string, required) - documents (file upload)

Requirements: - Validate all inputs - Store in database - Return provider ID + status

Example Response:

```
{
  "id": "provider_123",
  "status": "pending_verification",
  "message": "Registration successful. Awaiting verification."
}
```

2. GET /api/providers/:id

Function: Retrieve service provider details

Requirements: - Return provider details - Mask sensitive data appropriately (e.g., partial phone number, partial email)

Example Response:

```
{
  "id": "provider_123",
  "name": "ABC Provider",
  "email": "a***@example.com",
  "phone": "+94 77 *** **67",
  "business_type": "Service Operator",
}
```

```
"status": "pending_verification",
"created_at": "2026-01-21T10:00:00Z"
}
```

3. PUT /api/providers/:id/verify

Function: Admin endpoint to approve or reject service provider

Input: - status (string: “approved” or “rejected”) - notes (string, optional)

Requirements: - Authenticate as admin (JWT) - Update verification status - Return updated provider status

Example Response:

```
{
  "id": "provider_123",
  "status": "approved",
  "verified_at": "2026-01-21T14:30:00Z",
  "verified_by": "admin_user"
}
```

Technical Requirements

Language/Framework (choose one): - Python with Django or FastAPI - Node.js with Express

Database: - PostgreSQL

Must Include: - Proper input validation - Basic authentication using JWT - Comprehensive error handling - API documentation in README

Evaluation Criteria

Your submission will be evaluated on:

Criteria	Weight
Code Quality & Organization	25%
Clean, readable code with clear structure	
Security Considerations	20%
Input sanitization, proper auth, data protection	
Error Handling	15%

Graceful error responses, edge case handling

Database Design 15%

Appropriate schema, relationships, indexing

API Design Principles 15%

RESTful conventions, consistent responses

Documentation Clarity 10%

Clear README, setup instructions, API docs

Submission Instructions

Step 1: Create a Private GitLab Repository

1. Go to gitlab.com and sign in (or create a free account)
2. Create a **new private repository** for your submission
3. Name it something like: wii-group-backend-challenge

Step 2: Push Your Code

1. Include all source code
2. Include a comprehensive README with:
 - Project overview
 - Setup instructions (how to run locally)
 - Environment variables needed
 - API endpoint documentation
 - Any design decisions or assumptions
3. Include database schema/migrations
4. Include Postman collection OR OpenAPI/Swagger documentation

Step 3: Grant Us Access

1. Go to your repository's **Settings** → **Members**
2. Click **Invite members**
3. Add: careers@wijemannegroup.com
4. Set role: **Reporter** (read-only access)
5. Click **Invite**

Step 4: Notify Us

Reply to this email with: - Link to your GitLab repository - Any notes about your implementation - Confirmation that you've granted access

What We're Looking For

Strong submissions typically include:

- ✓ Clean, well-organized code structure
- ✓ Meaningful variable and function names
- ✓ Input validation with clear error messages
- ✓ Secure handling of sensitive data
- ✓ Proper use of environment variables (no hardcoded secrets)
- ✓ Database migrations included
- ✓ Clear README with setup instructions
- ✓ Thoughtful API responses
- ✓ Basic test coverage (bonus)

Red flags we watch for:

- ✗ Hardcoded credentials or secrets
- ✗ No input validation
- ✗ Poor error handling (generic errors, stack traces exposed)
- ✗ No documentation
- ✗ Disorganized code structure
- ✗ SQL injection vulnerabilities
- ✗ Missing authentication on protected routes

Timeline

- **Submission Deadline:** 48 hours from receiving this challenge
- **Review Period:** 1-2 business days after submission
- **Next Steps:** If successful, we'll schedule a technical interview to discuss your solution

Questions?

If you have questions about the requirements or need clarification:

- Email: careers@wijemannegroup.com
- We'll respond within a few hours during business hours

Note: Asking clarifying questions is encouraged and will not negatively impact your evaluation. We value candidates who seek clarity before building.

Important Notes

- This is your original work - do not copy from existing solutions
- You may use libraries and frameworks as needed

- Focus on quality over additional features
 - If you run out of time, document what you would have added
 - We respect your time - don't spend more than 4 hours on this
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Good luck! We look forward to reviewing your submission.

Wii Group

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